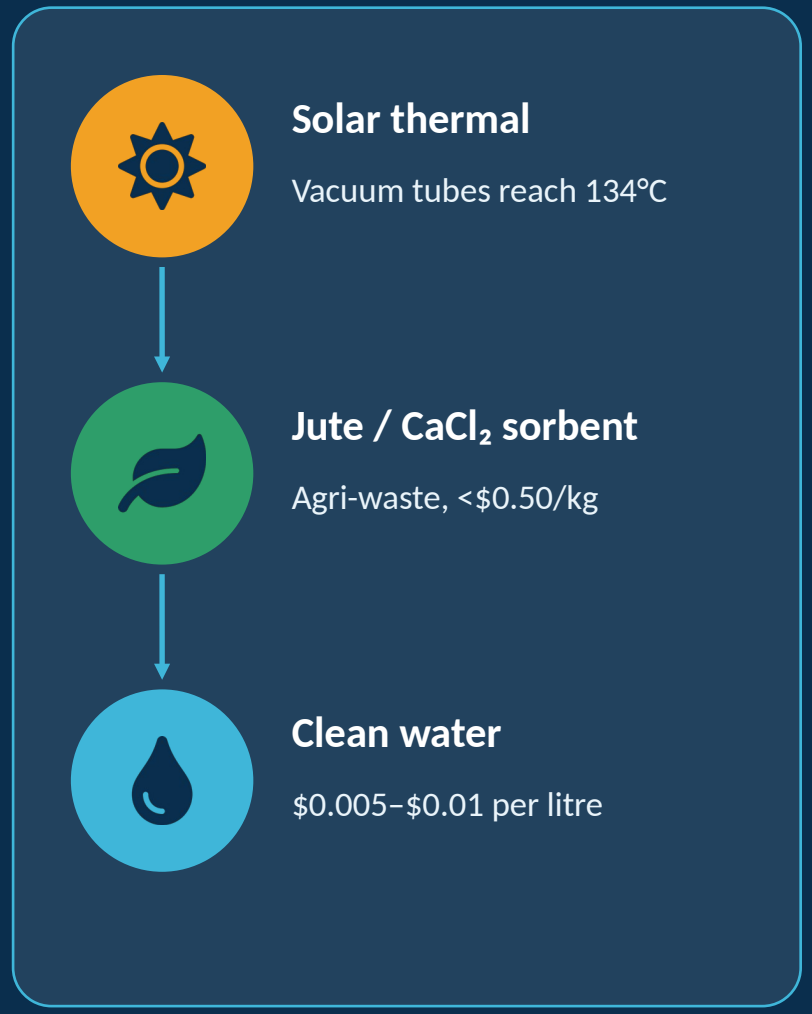
ACQUAHT

Drinking water from air at ₹0.40–₹0.80 per litre

We extract, filter and deliver potable water from atmospheric humidity using ultra-low-cost sorbents, solar thermal desorption and waste-heat recovery — at 1/20th the cost of bottled water.

We are not building another AWG. We are building the cheapest water source on Earth.



Raising $650,000 Seed | [Founder Name] | [Email] | [Phone] | [Website]

Safe water is scarce, expensive and logistically broken

2.2B

people lack safely managed drinking water

Indian cities approaching "Day Zero":

Bengaluru | Chennai | Delhi

What people pay today

Bottled

₹15–20/L

Tanker

₹5–10/L

Unreliable, often contaminated

Existing fixes fail

Desalination
Only works on the coast

Reverse osmosis (RO)
Needs a source water supply

Both fail where groundwater is depleted.

Demand already exists. Indian Railways has directed zonal railways, hospitals, staff colonies and remote level crossings to install atmospheric water generators (AWGs) where water scarcity exists.

The problem is not scarcity. It is the cost and logistics of delivering clean water.

Three forces make ultra-low-cost atmospheric water inevitable

Sorbent cost collapse

40–200x cheaper

Agri-waste and recycled-polymer sorbents cost $0.46–$2.50/kg vs $100+/kg for MOFs.

Free desorption energy

134°C from the sun

Solar thermal regenerates sorbents at zero marginal electricity. Photothermal hydrogels self-regenerate at 71°C under 0.5 sun.

Waste heat everywhere

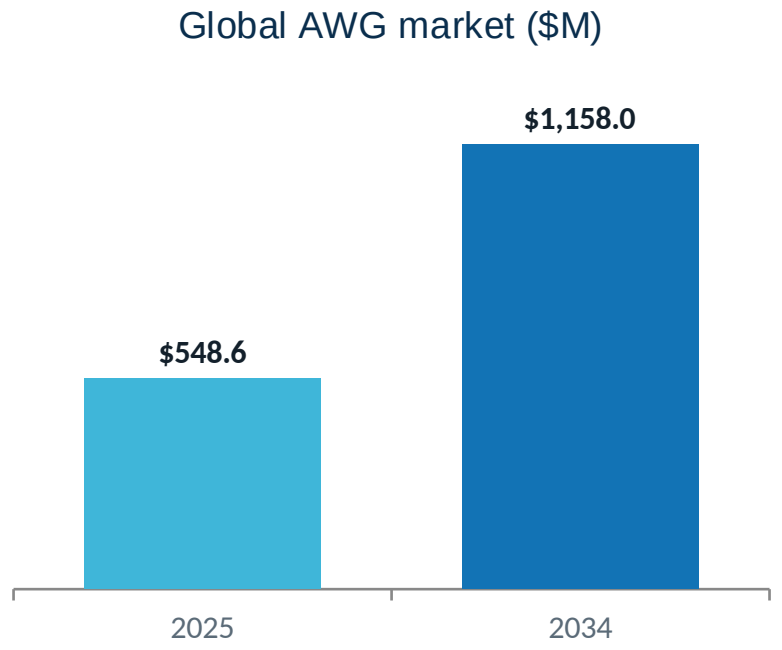
−34.4% energy

Hybrid systems hit 315 Wh/kg (50% RH, 30°C). Passive hybrid designs reach 0.2 kWh/L.

Policy tailwind

Govt. demand now

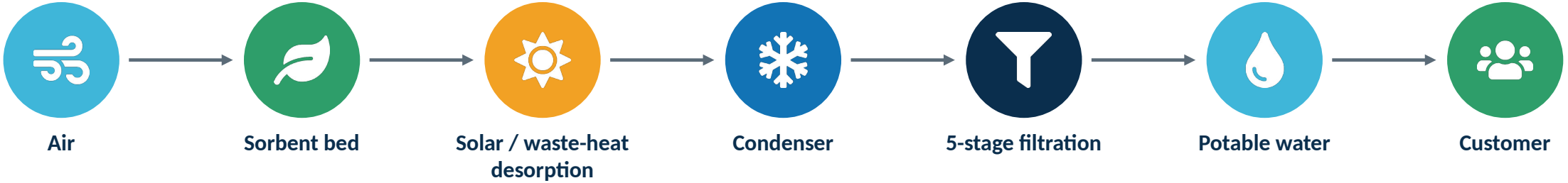
Indian Railways AWG mandate = immediate government procurement demand.



8.70% CAGR → $1.158B by 2034

The economics flipped. Water from air is no longer a premium product — it is the cheap one.

Hybrid sorption + solar thermal + waste heat + 5-stage filtration, as a service



01

Extract

Ultra-low-cost sorbent captures moisture from ambient air at 30–80% RH.

02

Regenerate

Solar thermal (134°C) and recovered waste heat release the vapour at near-zero marginal energy cost.

03

Condense & filter

5 stages: sediment, activated carbon, optional RO, UV sterilisation, mineralisation.

04

Deliver

Water-as-a-Service subscriptions, equipment leasing, franchise deployment.

05

Certify

ASSE 1090, BIS, NSF/ANSI 372, WHO Guidelines.

We do not sell machines. We sell certified water at a fraction of bottled-water cost.

$0.005–$0.01 per litre. Here is exactly how we get there

LCOW at scale

$0.005–$0.01

per litre

75–87% below conventional AWG ($0.02–$0.04/L)

= ₹0.40–₹0.80 per litre

COST DRIVER	CONVENTIONAL AWG		ACQUAHT	CUT
Sorbent material	~~MOF: $100+/kg~~	→	**Jute/$CaCl_2$ <$0.50/kg · PANSAP $2.50/kg · Starch-PEG/$CaCl_2$ $0.46/kg**	**40–200x**
Desorption energy	~~0.84–2.1 kWh/L electrical~~	→	**Solar thermal + waste heat: near-zero marginal**	**~90%**
Specific energy (system)	~~1.5–2.1 kWh/L~~	→	**0.2–0.5 kWh/L (passive / hybrid)**	**70–85%**
System CapEx (100 L/day)	~~$8,000–$12,000~~	→	**$2,030–$3,950 (BOM)**	**60–75%**
LCOW	~~$0.02–$0.04/L~~	→	**$0.005–$0.01/L**	**75–87%**

Two levers do almost all the work: sorbent cost and desorption energy. We attack both.

A four-stage hybrid system that wastes nothing

Stage 1

Condensation

Vapour compression makes water in humid air — and waste heat as a by-product.

waste heat →

Stage 2

Sorption

$CaCl_2$ salt-solution circuit uses Stage 1's waste heat to capture extra water with no major external energy.

+ solar heat →

Stage 3

Solar thermal desorption

Double-ended vacuum tubes (4.86 m², 134.2°C peak) drive daytime desorption at zero marginal electricity.

excess heat →

Stage 4

Thermal storage

Low-cost phase-change material or sand bed stores solar heat for night and cloudy days.

Stored heat drives overnight / cloudy-day desorption

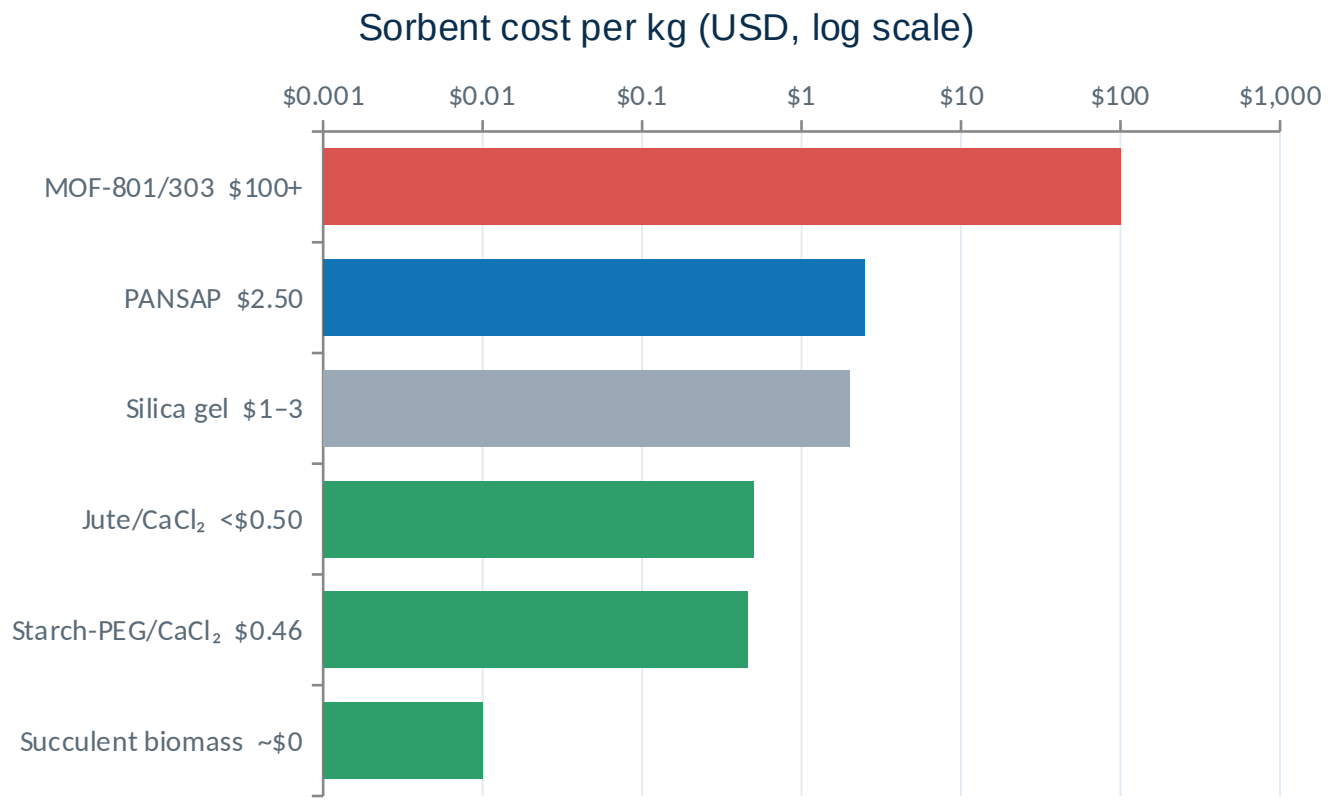
315 Wh/kg
hybrid operation

34.4%
energy saving vs basic vapour compression

0.2 kWh/L
passive configurations

We do not choose between condensation and sorption. We chain them so each stage powers the next.

We replaced $100/kg MOFs with agricultural waste and recycled polymer



MVP PROOF POINT

5,850 ml/day

from 12 kg of Jute/$CaCl_2$ composite at $0.086/L — published testing.

Beats silica gel

4–5x the water uptake of silica gel (0.26 g/g) — at a fraction of the cost.

The sorbent is the single biggest cost line in any AWG. We cut it by up to 200x — and it works better than silica gel.

Our sorbent line-up: tested, ranked and costed

Sorbent	Water uptake	Regeneration temp	Cost	Status
Jute/$CaCl_2$ composite	High at moderate RH	134°C (solar thermal)	$0.086/L water cost	MVP primary
Starch-PEG/$CaCl_2$ hydrogel	1.47 g/g	71.1°C at 0.5 sun	$0.46/kg	Parallel validation
Unprocessed succulent biomass	1.28 g/g (CR) · 1.08 g/g (AC)	65°C	~$0	Lab-validated
Agricultural straw waste	0.50–1.29 g/g	Solar-driven	Waste product	Lab-validated
PANSAP (recycled polymer)	Moderate–high at >40% RH	0.9–1.2 kWh/L	~$2.50/kg	Scale-up phase
Silica gel (baseline)	0.26 g/g	80–120°C	$1–3/kg	Reference only
MOF-801/303 (industry std.)	High at low RH	60–80°C	$100+/kg	Rejected on cost

Five sorbents beat silica gel on uptake or cost. We run the cheapest as primary and keep the rest in parallel.

We deleted the electricity bill from the desorption step

Solar thermal

Method	Temp	Source	Cost impact
Double-ended vacuum tube collector	134.2°C peak	Solar thermal	Zero electrical desorption energy
Photothermal hydrogel (MXene)	71.1°C at 0.5 sun	Passive solar	Self-regenerating under sunlight
Passive solar-driven desorption	Ambient	Passive	Near-zero energy cost
Thermal storage (PCM / sand)	Overnight release	Stored solar heat	Eliminates peak power demand

Waste-heat recovery

Configuration	Energy savings	Optimal conditions
Hybrid VCC + salt-solution circuit	34.4%	35–50% RH, 30°C
Condenser waste-heat recovery	Up to 67% less electricity	Various
Multicyclic sorption + waste heat	Desorption in <10 min	90°C waste heat
Passive hybrid with sorption	0.2 kWh/L achieved	Moderate humidity

Integration targets: data centres, industrial facilities, commercial buildings with HVAC — anywhere waste heat already exists.



Desorption is the energy hog in every sorption AWG. We made it free with sunlight and waste heat.

Water from air is pure at condensation. We make it potable and great-tasting

5-stage filtration

1. **Sediment filter (5–10 µm)**
   particulates
2. **Activated carbon block**
   VOCs, odours, chlorine
3. **Reverse osmosis (optional)**
   arid / dusty, higher TDS
4. **UV sterilisation (254 nm)**
   bacteria and viruses
5. **Mineralisation cartridge**
   Ca + Mg for taste and health

Testing protocol

Parameter	Standard	Method	Frequency
pH	6.5–8.5 (WHO)	IoT pH sensor	Daily
TDS	< 500 mg/L	IoT TDS sensor	Daily
Turbidity	< 5 NTU	Turbidity meter	Weekly
Microbial	0 CFU/100 mL	Membrane filtration	Monthly (3rd-party lab)
Heavy metals	Below WHO limits	ICP-MS	Quarterly (3rd-party lab)

Certifications

ASSE 1090 · **BIS (India)** · **NSF/ANSI 372** · **WHO Guidelines**

New sorbents: PANSAP condensate validated against Brazilian potable standards. Post-treatment with remineralisation is standard.

Cheap water is worthless if it is unsafe. Certification is a cost of entry, not an afterthought.

The MVP: two prototypes, one validation plan, a clear path to scale

Metric	MVP: 100 L/day	Scale-up: 1,000 L/day
Water output	100 L/day @ 30°C, 60% RH	1,000 L/day @ 30°C, 60% RH
Specific energy	< 1.5 kWh/L	< 1.0 kWh/L
Water quality	WHO/EPA potable	WHO/EPA potable
Uptime	> 90%	> 95%
Unit cost	< $8,000 all-in	< $40,000 all-in

Scale the same architecture 10x.

Prototype A

Condensation · humid climates

Hermetic rotary compressor (1.5 HP, R410A), microchannel evaporator + condenser coils, thermostatic expansion valve, 120–200 CFM axial fan, ESP32 IoT controller. Airflow 500–800 m^3/h; evaporator coil at 5–10°C.

Prototype B

Sorption · arid climates

Dual chamber, continuous: Chamber A adsorbs while Chamber B desorbs. 12 kg Jute/$CaCl_2$ sorbent. Solar thermal collector + waste-heat loop for desorption.

We prove the technology at 100 L/day for under $4,000 in materials — then scale the same architecture to 1,000 L/day.

Sorption MVP bill of materials: under $4,000

Component	Cost (USD)
Sorbent (Jute/$CaCl_2$, 12 kg)	$50–100
Adsorption chambers + heat exchangers	$400–800
Solar thermal collector (4.86 m^2)	$300–500
Condenser coil	$150–300
Fans	$80–150
Valves & controls	$200–400

Component	Cost (USD)
Thermal storage	$100–200
5-stage filtration	$300–600
Storage tank (100–200 L)	$100–200
IoT controller	$50–100
Enclosure & frame	$200–400
Miscellaneous	$100–200

TOTAL BOM

$2,030–$3,950

vs $8,000–$12,000 for a conventional 100 L/day AWG

Sorbent is just **$50–100** of it.

60–75% lower CapEx than a conventional AWG — and the costliest parts (chambers, filtration) get cheaper with volume.

MVP validation: lab → field → certification in 12 months



Design & simulation

Thermodynamic modelling, CAD, component sourcing

Delivers
Design package, BOM

Prototype build

Assemble condensation MVP and sorption MVP

Delivers
Two working prototypes

Lab testing

Controlled chamber: 20/25/30/35°C × 30–80% RH (6 steps)

Delivers
Performance curves, optimisation report

Field pilot

1 humid site (Mumbai/Chennai) + 1 arid site (Delhi/Ahmedabad), 3 months continuous

Delivers
Field data, MTBF, water quality reports

Certification

ASSE 1090, BIS, third-party water testing

Delivers
Certifications, validated unit economics

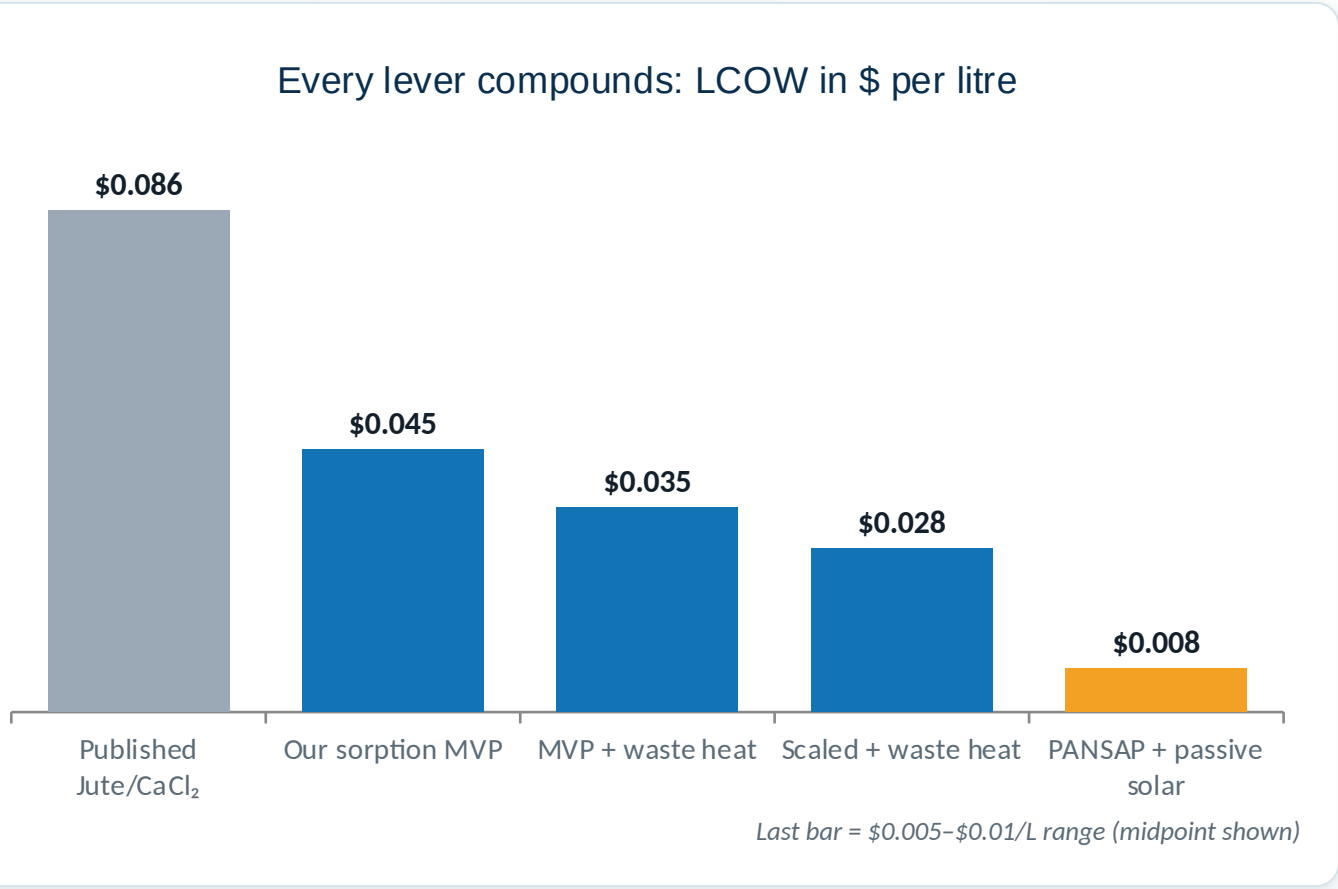
Scale-up design

Design 1,000 L/day unit from MVP learnings

Delivers
Scale-up design, manufacturing plan

We do not claim performance. We measure it: L/h, kWh/L, TDS and pH, and mean time between failures.

Under 24-month payback. 70–85% gross margin at scale



<24 mo
payback

70–85%
gross margin at scale

₹5–8/L
price in India ($0.06–$0.10)

LCOW by climate (our system)

Climate	Grid	Existing solar	Solar thermal + waste heat
Tropical	$0.06/L	$0.02/L	**$0.008/L**
Temperate	$0.09/L	$0.04/L	**$0.010/L**
Arid	$0.17/L	$0.09/L	**$0.012/L**

Every cost-reduction lever compounds. Sorbent + solar thermal + waste heat takes us from competitive to unbeatable.

Unit economics in detail

Assumptions: 100 L/day unit · 300 days/year = 30,000 L/year · 5-year CapEx amortisation

Parameter	Conventional AWG	Our sorption MVP	Our scaled system
CapEx	$8,000–12,000	$4,000	$3,500 (at volume)
Annual production	30,000 L	30,000 L	30,000 L
Amortised CapEx / year	$1,600–2,400	$800	$700
Sorbent replacement	—	$50/yr (2-yr cycle)	$25/yr (PANSAP, 3-yr cycle)
Filters	$300/yr	$300/yr	$250/yr
Labour	$200/yr	$200/yr	$150/yr
Electricity	$1,200–2,500/yr	$0–200/yr	$0–100/yr
Total annual OpEx	$3,300–5,400	**$1,350**	**$1,125**
LCOW	$0.06–$0.09/L	**$0.045/L**	**$0.037/L**
LCOW with waste-heat recovery (−34%)	—	**$0.035/L**	**$0.028/L**
LCOW with PANSAP + passive solar	**—**	**—**	**$0.005–0.01/L**

Electricity is the line we delete: $1,200–2,500/yr for a conventional AWG vs $0–200/yr for ours.

A billion-dollar market with a government-backed beachhead

TAM $1.158B

Global AWG market by 2034 · 8.70% CAGR

SAM $250M+

India + UAE + Philippines + Africa institutional water

SOM $8M

2,000 units by Year 3

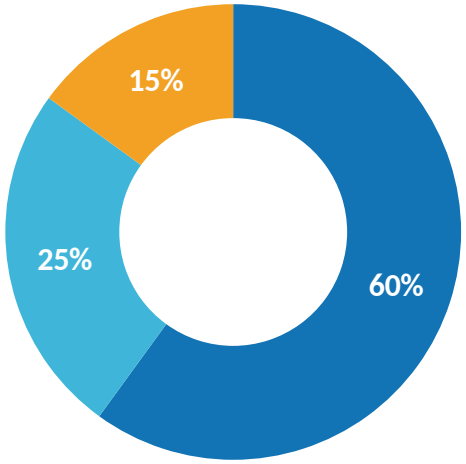
Segment	Water need	Willingness to pay	Priority
Railway stations (India)	500–5,000 L/day	₹3.5–5/L	Tier 1
Hospitals & clinics	200–2,000 L/day	₹5–8/L	Tier 1
Corporate campuses	1,000–10,000 L/day	₹4–6/L	Tier 2
Data centres (waste-heat sites)	5,000–50,000 L/day	$0.10–0.20/L	Tier 2
Residential (off-grid)	10–50 L/day	$0.20–0.40/L	Tier 3
Events & hospitality	100–1,000 L/day	₹8–15/L	Tier 3

Why data centres: they make continuous waste heat and already pay for cooling water. We turn their waste heat into free desorption energy.

Start with Indian Railways tenders. Expand into waste-heat-rich facilities where our economics are strongest.

Four revenue streams. One recurring-margin engine

Revenue mix



60% WaaS — monthly fee for guaranteed water volume
25% Sale or lease-to-own, 15%+ ROI
15% Local entrepreneurs deploy and maintain
+ Premium bottled water (optional): hotels, airlines, events

Price per litre in India

Bottled
₹15–20/L

Tanker
₹5–10/L

AcquaHT
₹3.5–8/L

Undercuts bottled water by 60–80%.
UAE / Saudi: $0.10–$0.15/L — below desalinated delivery in remote areas.

NEW: Sorbent-as-a-Component

License our Jute/$CaCl_2$ and PANSAP formulations to existing AWG makers as a drop-in efficiency upgrade. Follows the EU-funded ThinAirWater model. High-margin, capital-light — monetises our IP independent of unit sales.

We own the distribution and the sorbent IP. Both generate recurring revenue.

Beachhead: Indian Railways. Then waste-heat sites, hospitals, franchises

Phase 1 · Beachhead

Months 0–9

- Bid for Indian Railways AWG tenders: zonal railways, hospitals, colonies, level crossings
- 10-unit pilot in Mumbai/Chennai (humid) + Delhi (arid)
- Publish validated LCOW + water quality report

Phase 2 · Scale

Months 9–18

- Hospitals, corporate campuses, data centres with existing waste heat
- Sign 3 franchise partners
- Break-even on pilot units

Phase 3 · Expand

Months 18–30

- Enter UAE and Philippines
- PANSAP sorbent line for arid India
- Start sorbent-as-a-component licensing
- Raise Series A

Sales funnel

Tender → Pilot → Scale → Franchise → License

Channels: government tenders · direct enterprise sales · franchise network · OEM sorbent licensing

Tenders give volume. Waste-heat sites give margins. Franchising gives speed. Licensing gives capital-light upside.

Traction & validation: [insert current traction here]

Pipeline

[X] letters of intent from hospitals, corporates or data centres

Government

[Y] railway tender submitted or pre-qualified

Prototypes

Condensation + sorption MVP build status: [status]

Sorbent validation

Jute/$CaCl_2$ and PANSAP lab performance data: [data]

Certifications

ASSE 1090 and BIS applications: [in progress]

Partnerships

[HVAC manufacturer], [solar thermal supplier], [waste-heat host site]

Water quality

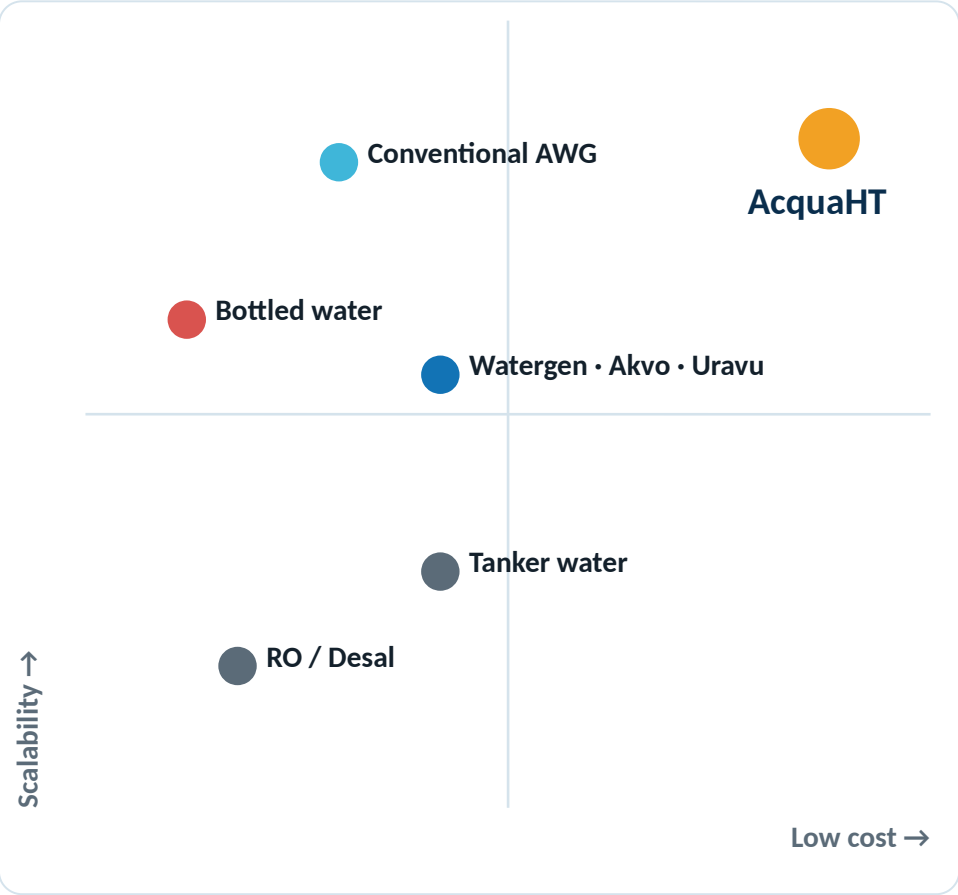
Third-party test results from pilot units: [results]

Next milestone

[Insert the next proof point and its date]

We are not starting from zero — we have a validation plan with measurable milestones.

We win on cost and defend on sorbent IP and system architecture



Competitor	Weakness	Our advantage
Bottled water	₹15–20/L, plastic waste	₹0.40–0.80/L production cost, zero plastic
Tanker water	Unreliable, contaminated	On-site, certified, IoT-monitored
RO / Desalination	Needs source water, coastal only	Works anywhere RH >30%
Conventional AWG	$0.06–0.17/L, high CapEx	$0.005–0.01/L, 60–75% lower CapEx
Watergen, Akvo, Uravu	Strong brands, MOF/silica-based	Agri-waste sorbents, solar thermal, waste heat

Anyone can buy a compressor. Nobody else has our sorbent cost structure or our hybrid thermal architecture.

Our moat has three layers

1. Sorbent formulation IP

Jute/$CaCl_2$ composite and PANSAP processing know-how. Patentable, hard to reverse-engineer and licensable.

2. Hybrid system architecture

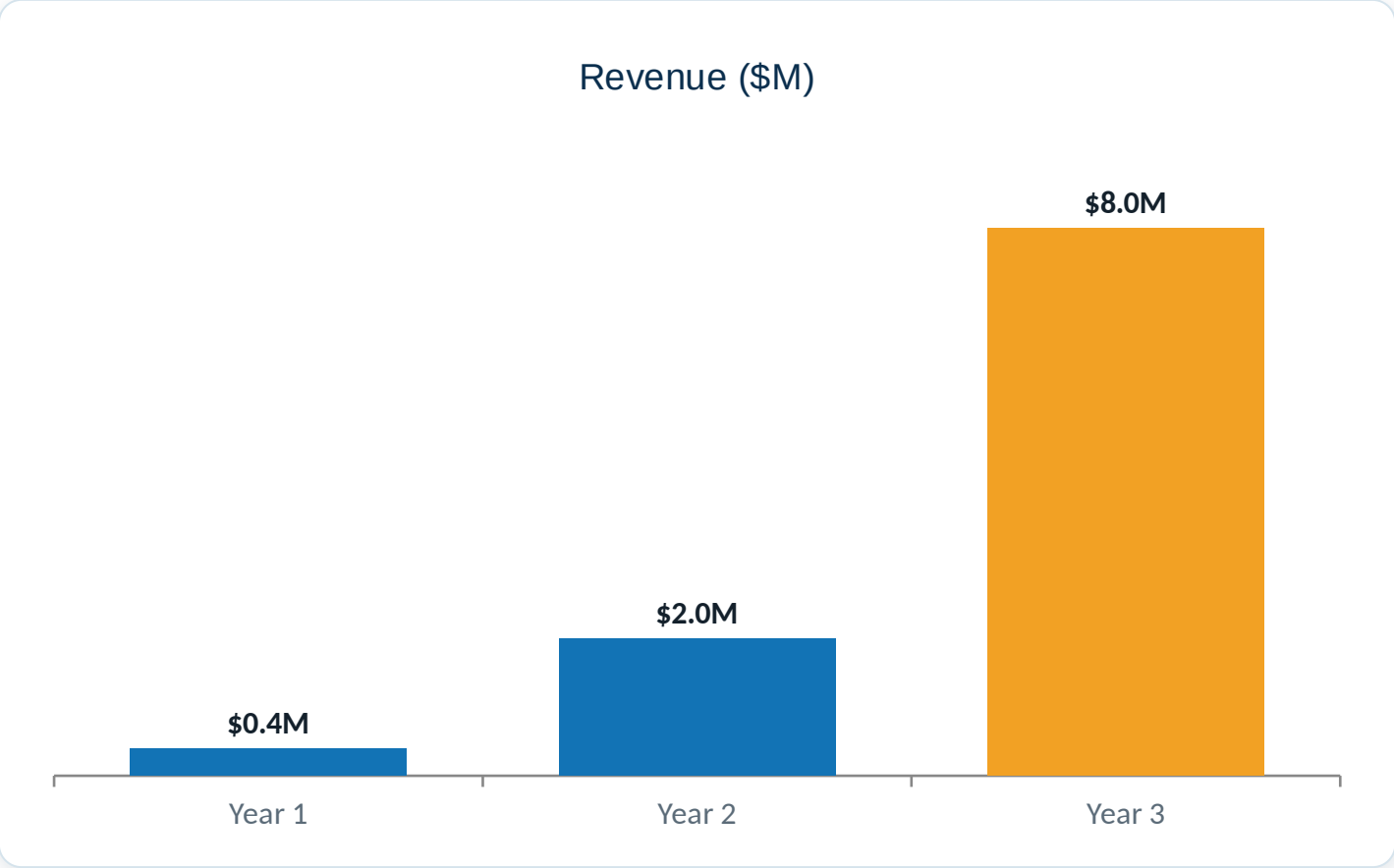
The four-stage condensation → sorption → solar thermal → storage chain, including the waste-heat recovery loop. Patentable as a system claim.

3. Deployment data

Climate-specific performance curves from lab and field testing. Every unit sold improves our optimisation models.

Cost is our attack. IP, architecture and data are our defence.

$8M revenue by Year 3. Break-even by Month 18–20



Year	Units	Revenue	Net margin
Year 1	100	$400,000	−15%
Year 2	500	$2,000,000	15%
Year 3	2,000	$8,000,000	30%+

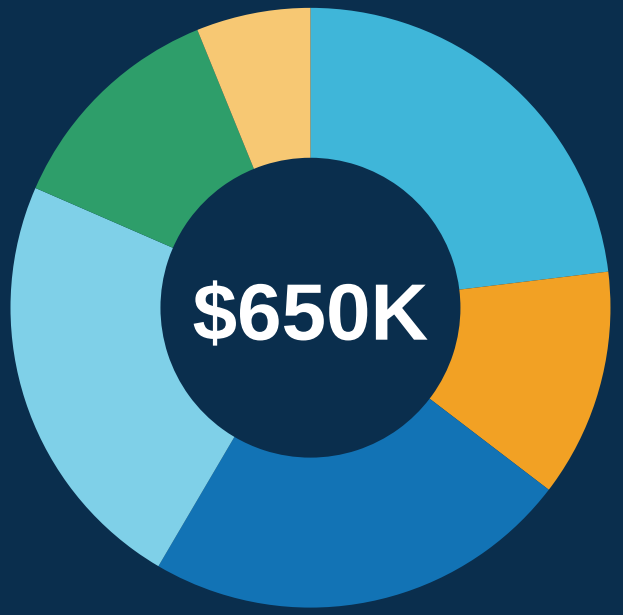
Blended model: 60% WaaS, 25% equipment, 15% franchise/licensing.

Avg revenue per unit: $4,000/year.

Margin grows with: falling sorbent costs and the shift to PANSAP at volume.

Year 1 loses money on purpose: it buys the proof. Margins expand as sorbent costs fall.

Raising $650,000 to prove the tech and ship the first units



MVP build, lab testing & field pilot (2 prototypes, 2 sites)	$150,000	23%
Certification (ASSE 1090, BIS, WHO validation)	$80,000	12%
Manufacturing partnership & tooling	$150,000	23%
Pilot deployment (25–50 units, humid + arid)	$150,000	23%
Working capital	$80,000	12%
Marketing, sales & government tenders	$40,000	7%

Milestones: validated MVPs (M12) → break-even (M18–20) → Series A (M18–30)

This round buys

- Two validated 100 L/day MVPs
- Field data: humid + arid
- ASSE 1090 + BIS certification
- 25–50 deployed units
- Break-even unit economics proven
- Series A readiness

This round funds the technology proof and the first commercial deployments. Series A scales it.

Operators who can build, validate and scale

CEO
[Name]
Water, energy or operations background

CTO
[Name]
Thermodynamics, HVAC/refrigeration or materials science

COO
[Name]
Supply chain, franchise operations or government sales

Head of Materials
Key hire
Sorbent chemistry and composite formulation

Advisors

- Water policy expert (India regulation & tenders)
- Solar thermal engineer
- Certification consultant (ASSE 1090 / BIS)
- Waste-heat integration specialist (industrial & data centre)

Hiring plan (funded by this round)

- 2 engineers (AWG systems, IoT)
- 1 materials scientist (sorbent formulation)
- 1 government sales lead
- 1 franchise manager

We have the thermodynamic depth, the materials expertise and the government sales experience to execute.

From two prototypes to 2,000 units in 48 months



Phase	Months	Milestones	LCOW
Foundation	0–3	Incorporate; secure pilot sites; file ASSE 1090; source sorbents	—
MVP & Pilot	3–12	2 prototypes; lab + field validation; certifications; validate $0.04/L	$0.045
Cost Reduction I	6–12	Solar thermal collector; waste-heat recovery loop	$0.03
Scale-Up	12–18	100 units; 3 franchise partners; break-even	$0.02
Cost Reduction II	15–24	PANSAP; modular hydrocell; photothermal MXene	$0.015
Expansion	18–30	UAE, Philippines; sorbent licensing; Series A	$0.01
Cost Reduction III	24–36	Thermal storage for overnight desorption; passive hybrid	$0.005–0.01
Scale	30–48	2,000+ units deployed	$0.005

Every phase has a cost target, not just a volume target. Cost reduction and scale happen in parallel.

We have a plan for every major risk

Risk	Impact	Mitigation
Low humidity in arid markets	High	Sorption MVP validated at 30% RH in lab; sorption is the primary tech, not a fallback
Water quality from new sorbents	High	PANSAP condensate validated vs potable standards. 5-stage filtration + remineralisation. Weekly testing
Sorbent degradation over cycles	Medium	Jute/$CaCl_2$ is chemically stable; PANSAP shows cycling stability. Test 1,000+ cycles in lab
Solar thermal variability	Medium	PCM / sand-bed thermal storage for night and cloudy days. Hybrid with waste heat
Waste-heat availability	Medium	Prioritise sites with HVAC or industrial waste heat. Baseline economics do not depend on it
Component supply chain	Medium	Multi-source. Prioritise Indian HVAC makers (Godrej, Kirloskar)
Regulatory ban on AWG	Medium	Position as “Monsoon Resilience Units”. Engage policy makers early
Competition from incumbents	Medium	Defend on sorbent IP and hybrid architecture patents. Move fast on certification
Technology obsolescence	Low	Roadmap already targets the theoretical minimum. Hydrogel is upside, not a dependency



We are not betting on one technology, one climate or one revenue stream.

Water access, less plastic, carbon credits and heat reuse

1 unit

100 L/day

→

30,000 L

per year

→

~60,000

500 ml bottles avoided

→

[X] t CO_2

avoided per year

Energy

We use heat that would be vented — a net efficiency gain for host sites.

Water access

Off-grid communities, hospitals, railway stations, remote level crossings.

Carbon credits

Solar-thermal-powered AWG generates verifiable credits.

UN SDGs

SDG 6 Clean Water · SDG 7 Clean Energy · SDG 13 Climate Action

A profitable business with measurable social and environmental returns — that makes host facilities more efficient, not less.

ACQUAHT

We turn waste heat and sunlight into drinking water.

$0.005–0.01

per litre at scale

1/20th

the cost of bottled water

$650K

seed round

[Founder Name] · [Email] · [Phone] · [Website]

Appendix

- Detailed BOM and cost per unit (condensation + sorption)
- LCOW methodology and full assumption set
- Sorbent data: Jute/$CaCl_2$, PANSAP, starch-PEG/$CaCl_2$, succulent biomass, straw
- Solar thermal and waste-heat integration schematics
- Lab test protocol and field pilot design
- Certification path: ASSE 1090, BIS, WHO, NSF/ANSI 372
- Pilot water quality data
- Competitive matrix and patent landscape summary
- Franchise agreement summary and sorbent licensing term sheet
- Full financial model (Excel on request)

Every claim in this deck is backed by a testable assumption or a published result.

One-page executive summary

	AcquaHT
What	Atmospheric water from ultra-low-cost agri-waste and recycled-polymer sorbents, solar thermal desorption and waste-heat recovery, plus 5-stage filtration.
Cost	$0.005–$0.01/L at scale (₹0.40–₹0.80/L). 75–87% below conventional AWG. 60–80% below bottled water.
Where	India first (Indian Railways, hospitals, corporates, data centres), then UAE, Philippines, Africa.
Tech	Hybrid 4-stage: condensation → sorption → solar thermal → thermal storage. Jute/$CaCl_2$ and PANSAP at $0.46–$2.50/kg. 134°C collectors. 34.4% saving from waste heat.
MVP	Two 100 L/day prototypes (condensation + sorption), validated in lab and field, scaling to 1,000 L/day.
Model	60% WaaS · 25% equipment sales · 15% franchise + sorbent licensing.
Traction	[Insert LOIs, prototype status, certifications]
Financials	Y1 $400K · Y2 $2M · Y3 $8M. Break-even M18–20. Gross margin 70–85% at scale.
Ask	$650,000 Seed · Contact: [Name] · [Email] · [Phone] · [Website]